

1. Gender Distribution (Pie Chart)

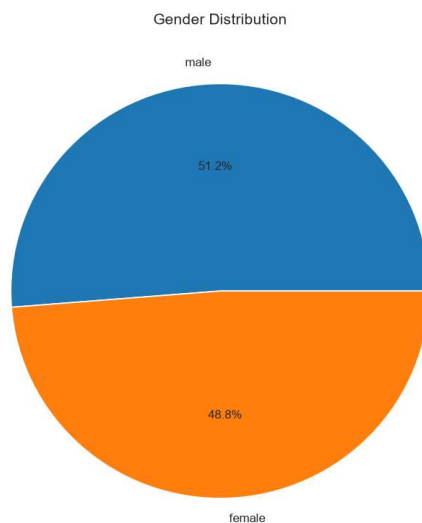
Description

The Gender Distribution pie chart provides a visual representation of the proportion of male and female participants included in the Teen Mental Health dataset. Examining the gender composition is an essential step in Exploratory Data Analysis (EDA) because it helps determine whether the dataset is balanced and representative. A balanced dataset ensures that any conclusions drawn from subsequent analyses are less likely to be influenced by sampling bias.

Code

```
plt.figure(figsize=(6,6))  
df['gender'].value_counts().plot(kind='pie', autopct='%1.1f%%')  
plt.title('Gender Distribution')  
plt.ylabel('')  
plt.show()
```

Graph



Interpretation

The pie chart indicates the percentage of male and female students represented in the dataset. The proportions suggest that both genders are well represented, allowing meaningful comparative analysis in later stages of the project. A balanced gender distribution increases the credibility of statistical observations and minimizes the possibility of gender bias affecting the overall results. This visualization establishes the demographic structure of the dataset and confirms that it is suitable for analyzing behavioural and mental health characteristics across different gender groups. Future analyses may explore whether stress levels, anxiety levels, sleep duration, or social media addiction differ between male and female students.